Lecture Notes

Lecture 7

Inner Product Spaces

So far our vector spaces have had no geometry: no length, no angle, no notion of two vectors being perpendicular. The *inner product* supplies all three at once, and with it linear algebra becomes the language of quantum mechanics. The overlap $\langle \phi \mid \psi \rangle$ is a probability amplitude; the normalization $\langle \psi \mid \psi \rangle = 1$ fixes total probability; orthogonality $\langle \phi \mid \psi \rangle = 0$ means two states are perfectly distinguishable. This lecture sets up inner products in the *physics convention* and in Dirac's bra–ket notation, and proves the inequality that underlies every amplitude bound in the subject: Cauchy–Schwarz.

A word on convention. We take the inner product to be *conjugate-linear in the first slot and linear in the second*—the choice universal in physics, so that $\langle \phi \mid \psi \rangle$ is linear in the ket $|\psi\rangle$. Many mathematics texts put the conjugation in the second slot; every formula converts by swapping the two arguments.

Learning objectives

By the end of this lecture you will be able to:

- Verify the inner-product axioms in the physics convention (conjugate-linear first slot).
- Move fluently between $\langle \cdot, \cdot \rangle$ and Dirac $\langle \cdot \mid \cdot \rangle$ notation.
- Use norms, orthogonality, and the Pythagorean theorem.
- Prove and apply Cauchy–Schwarz and the triangle inequality; use polarization.

1 Inner products

Definition 7.1 (Inner product, physics convention). An *inner product* on a complex vector space V assigns to each ordered pair $u, v \in V$ a scalar $\langle u, v \rangle \in \mathbb{C}$ such that

1. *conjugate symmetry*: $\langle u, v \rangle = \overline{\langle v, u \rangle}$;
2. *linearity in the second slot*: $\langle u, av + bw \rangle = a\langle u, v \rangle + b\langle u, w \rangle$;
3. *positive definiteness*: $\langle v, v \rangle \geq 0$, with equality iff $v = 0$.

A vector space with an inner product is an *inner product space*.

Conjugate symmetry forces $\langle v, v \rangle = \overline{\langle v, v \rangle}$ to be real, so condition (3) makes sense. Combining (1) and (2) gives *conjugate-linearity in the first slot*:

$$\langle au + bw, v \rangle = \overline{\langle v, au + bw \rangle} = \overline{a\langle v, u \rangle + b\langle v, w \rangle} = \overline{a}\,\langle u, v \rangle + \overline{b}\,\langle w, v \rangle.$$

The scalars come out of the first slot conjugated and out of the second slot unchanged—"one and a half" times linear, or *sesquilinear*. Over $\mathbb{R}$ the conjugations vanish and an inner product is just a symmetric, positive-definite bilinear form.

Example 7.2 (The standard inner products). On $\mathbb{C}^n$, the standard inner product is

$$\langle u, v \rangle = \sum_{k=1}^{n} \overline{u_k}\, v_k = u^\dagger v,$$

conjugating the *first* vector's components (here $u^\dagger = \overline{u}^{\mathsf{T}}$ is the conjugate transpose). On $\mathbb{R}^n$ this is the ordinary dot product $\sum_k u_k v_k$. On the space of continuous functions on $[a, b]$,

$$\langle f, g\rangle = \int_a^b \overline{f(t)}\, g(t)\, dt$$

is an inner product—the model for the state spaces of quantum mechanics. ◁

Example 7.3 (A computation in $\mathbb{C}^2$). With $u = (1, i)$ and $v = (2,\ 1 - i)$ in $\mathbb{C}^2$,

$$\langle u, v\rangle = \overline{1} \cdot 2 + \overline{i} \cdot (1 - i) = 2 + (-i)(1 - i) = 2 + (-i - 1) = 1 - i,$$

while $\langle v, u\rangle = \overline{2} \cdot 1 + \overline{(1 - i)} \cdot i = 2 + (1 + i)i = 2 + (i - 1) = 1 + i = \overline{\langle u, v\rangle}$, as conjugate symmetry demands. ◁

Example 7.4 (The Frobenius inner product). On $M_n(\mathbb{C})$, $\langle A, B\rangle = \mathrm{tr}(A^\dagger B) = \sum_{j,k} \overline{A_{jk}}\, B_{jk}$ is an inner product; it is the standard inner product on $\mathbb{C}^{n^2}$ in disguise, and it will measure the "size" of operators later. ◁

Example 7.5 (Weighted products and a non-example). For fixed weights $w_1, \ldots, w_n > 0$, $\langle u, v\rangle = \sum_k w_k \overline{u_k} v_k$ is again an inner product on $\mathbb{C}^n$. Positivity is exactly what requires $w_k > 0$: with $w = (1, -1)$ the nonzero vector $(0, 1)$ would have $\langle v, v\rangle = -1 < 0$, so the form fails to be an inner product. The positivity axiom is what makes $\|v\| = \sqrt{\langle v, v\rangle}$ a genuine length. ◁

Proposition 7.6 (Basic facts). In any inner product space, $\langle v, 0\rangle = \langle 0, v\rangle = 0$, and if $\langle u, x\rangle = \langle w, x\rangle$ for every x, then $u = w$.

Proof. $\langle v, 0\rangle = \langle v, 0 \cdot 0\rangle = 0 \cdot \langle v, 0\rangle = 0$, and $\langle 0, v\rangle = \overline{\langle v, 0\rangle} = 0$. For the second claim, conjugate-linearity in the first slot gives $\langle u - w, x\rangle = 0$ for all x; taking $x = u - w$ yields $\|u - w\|^2 = 0$, so $u = w$. ■

Example 7.7 (Checking the axioms). For the standard product on $\mathbb{C}^n$: conjugate symmetry is $\langle u, v\rangle = \sum_k \overline{u_k} v_k = \overline{\sum_k \overline{v_k} u_k} = \overline{\langle v, u\rangle}$; linearity in the second slot is the distributive law; and $\langle v, v\rangle = \sum_k |v_k|^2 \geq 0$ vanishes only when every $v_k = 0$. The same three checks, with sums replaced by integrals, verify the L^2 product of Example 7.2. ◁

Remark 7.8 (Why the conjugation is not optional). Over $\mathbb{C}$, conjugate symmetry forces $\langle v, v\rangle$ to be real, so the requirement $\langle v, v\rangle \geq 0$ even makes sense; without the conjugation in the first slot, $\langle v, v\rangle$ could be complex and "length" would be undefined. The physics convention keeps amplitudes linear in the ket while guaranteeing real, positive norms.

2 Dirac notation

Physicists write vectors as *kets* $|\psi\rangle$ and pair them using *bras*. To each vector ϕ Dirac associates the bra $\langle\phi|$, the linear functional that sends $|\psi\rangle$ to the inner product:

$$\langle\phi|\colon |\psi\rangle \longmapsto \langle\phi \mid \psi\rangle = \langle\phi, \psi\rangle.$$

Because the inner product is conjugate-linear in its first slot, the correspondence $|\phi\rangle \mapsto \langle\phi|$ is conjugate-linear: $a|\phi\rangle + b|\chi\rangle \mapsto \overline{a}\,\langle\phi| + \overline{b}\,\langle\chi|$. The bracket $\langle\phi \mid \psi\rangle$ is then linear in the ket and conjugate-linear in the bra—exactly Definition 7.1.

The reversed product $|\psi\rangle\langle\phi|$ is an *operator*: it sends $|\chi\rangle$ to $|\psi\rangle\langle\phi \mid \chi\rangle$, a scalar multiple of $|\psi\rangle$. Such rank-one "outer products" build projections and, in Lecture 8, resolutions of the identity.

Remark 7.9 (The Born rule). For normalized states, $\langle \phi \mid \psi \rangle$ is the *probability amplitude* to find a system prepared in $|\psi\rangle$ to be in $|\phi\rangle$, and $|\langle \phi \mid \psi \rangle|^2$ is the probability. Cauchy–Schwarz below will show this probability never exceeds 1.

Example 7.10 (An amplitude). Let $|\psi\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big)$ and $|\phi\rangle = |0\rangle$ in an orthonormal two-state basis $|0\rangle, |1\rangle$ (so $\langle 0 \mid 0\rangle = \langle 1 \mid 1\rangle = 1$, $\langle 0 \mid 1\rangle = 0$). Then $\langle \phi \mid \psi\rangle = \frac{1}{\sqrt{2}}\big(\langle 0 \mid 0\rangle + \langle 0 \mid 1\rangle\big) = \frac{1}{\sqrt{2}}$, so the probability of finding the system in $|0\rangle$ is $|\langle \phi \mid \psi\rangle|^2 = \frac{1}{2}$. ◁

Example 7.11 (An outer product as a matrix). With $|0\rangle = (1, 0)$ and $|1\rangle = (0, 1)$, the outer product

$$|0\rangle\langle 0| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

is the projection onto $|0\rangle$, and $|0\rangle\langle 0| + |1\rangle\langle 1| = I$ is a *resolution of the identity*. Projections and observables will be assembled from such outer products in the next lectures. ◁

Remark 7.12 (Expansions and expectation values). In an orthonormal basis $\{|n\rangle\}$ the *completeness relation* $\sum_n |n\rangle\langle n| = I$ expands any state as $|\psi\rangle = \sum_n |n\rangle\langle n \mid \psi\rangle$, with coefficients the amplitudes $\langle n \mid \psi\rangle$. The *expectation value* of an observable A in the state $|\psi\rangle$ is the bra–ket sandwich $\langle A\rangle = \langle \psi | A | \psi\rangle$, which Lecture 9 will show is real for observables. We construct such bases systematically in Lecture 8.

Example 7.13 (An expectation value). For $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and the observable $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$, since $\sigma_z|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$,

$$\langle \sigma_z\rangle = \langle \psi | \sigma_z | \psi\rangle = \tfrac{1}{2}\big(\langle 0 \mid 0\rangle - \langle 0 \mid 1\rangle + \langle 1 \mid 0\rangle - \langle 1 \mid 1\rangle\big) = \tfrac{1}{2}(1 - 1) = 0,$$

the average of the outcomes $+1$ and -1, each with probability $\frac{1}{2}$. The bra–ket sandwich computes averages directly. ◁

3 Norm, orthogonality, and the Pythagorean theorem

Definition 7.14 (Norm and orthogonality). The *norm* of v is $\|v\| = \sqrt{\langle v, v\rangle}$. A vector with $\|v\| = 1$ is a *unit vector* (a *normalized state*). Two vectors are *orthogonal*, $u \perp v$, if $\langle u, v\rangle = 0$.

The norm is homogeneous, $\|av\| = \sqrt{\langle av, av\rangle} = \sqrt{\bar{a}\, a\, \langle v, v\rangle} = |a|\,\|v\|$, and vanishes only at 0. The single most useful identity relating norm and orthogonality is the following.

Theorem 7.15 (Pythagorean theorem). If $u \perp v$, then $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.

Proof. Expanding with sesquilinearity,

$$\|u + v\|^2 = \langle u + v, u + v\rangle = \|u\|^2 + \langle u, v\rangle + \langle v, u\rangle + \|v\|^2.$$

Orthogonality kills the cross terms $\langle u, v\rangle = \langle v, u\rangle = 0$, leaving $\|u\|^2 + \|v\|^2$. ∎

Example 7.16 (Normalizing a state). The vector $|\psi\rangle = (1, i, 1) \in \mathbb{C}^3$ has $\|\psi\|^2 = \langle \psi, \psi\rangle = |1|^2 + |i|^2 + |1|^2 = 3$, so the normalized state is $\frac{1}{\sqrt{3}}(1, i, 1)$, a unit vector describing the same physics. The induced distance $\|u - v\|$ makes V a metric space—the setting for convergence and completeness in Lectures 13–14. ◁

Definition 7.17 (Orthogonal complement). The *orthogonal complement* of a subspace $U \subseteq V$ is $U^\perp = \{v \in V : \langle u, v\rangle = 0 \text{ for all } u \in U\}$, itself a subspace of V.

Proposition 7.18 (Pythagoras for many vectors). If $v_1, \dots, v_m$ are pairwise orthogonal, then $\|v_1 + \cdots + v_m\|^2 = \|v_1\|^2 + \cdots + \|v_m\|^2$.

Proof. Expand $\langle \sum_j v_j, \sum_k v_k\rangle = \sum_{j,k}\langle v_j, v_k\rangle$; every off-diagonal term vanishes by orthogonality, leaving $\sum_j \|v_j\|^2$. ■

Example 7.19 (Pythagoras with three components). The standard basis vectors of $\mathbb{R}^3$ are pairwise orthogonal, so for $v = (2, -1, 2) = 2e_1 - e_2 + 2e_3$, Proposition 7.18 gives $\|v\|^2 = 2^2 + (-1)^2 + 2^2 = 9$ and $\|v\| = 3$—the Euclidean length read as a sum over orthogonal components, the same bookkeeping that makes $\|\psi\|^2 = \sum_n |\langle n \mid \psi\rangle|^2$ a total probability. ◁

Example 7.20 (Independence via the Gram matrix). The *Gram matrix* of $v_1, \dots, v_m$ has entries $G_{jk} = \langle v_j, v_k\rangle$. The vectors are linearly independent if and only if G is invertible: a dependence $\sum_k c_k v_k = 0$ yields $Gc = 0$ after pairing with each v_j, and conversely. For $v_1 = (1, 0)$, $v_2 = (1, 1)$ in $\mathbb{R}^2$, $G = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ has $\det G = 1 \neq 0$, confirming independence. ◁

Example 7.21 (An orthonormal pair). In $\mathbb{C}^2$ the vectors $e_1 = \frac{1}{\sqrt{2}}(1, 1)$ and $e_2 = \frac{1}{\sqrt{2}}(1, -1)$ have $\langle e_1, e_1\rangle = \langle e_2, e_2\rangle = 1$ and $\langle e_1, e_2\rangle = \frac{1}{2}(1 - 1) = 0$: they are orthonormal. Any $u = (a, b)$ then expands as $u = \langle e_1, u\rangle\, e_1 + \langle e_2, u\rangle\, e_2$, the prototype of the orthonormal expansions of Lecture 8. ◁

Example 7.22 (Norms and overlaps of functions). On $C[0, 1]$ with $\langle f, g\rangle = \int_0^1 \overline{f}\, g\, dt$, the function $f(t) = t$ has $\|f\|^2 = \int_0^1 t^2\, dt = \frac{1}{3}$, so $\|f\| = 1/\sqrt{3}$. The functions 1 and t are not orthogonal, since $\langle 1, t\rangle = \int_0^1 t\, dt = \frac{1}{2} \neq 0$; Gram–Schmidt (Lecture 8) will replace them by an orthogonal pair, the first Legendre polynomials. ◁

4 The Cauchy–Schwarz inequality

Every vector splits into a part along a given direction and a part orthogonal to it; that decomposition proves the central inequality of the subject.

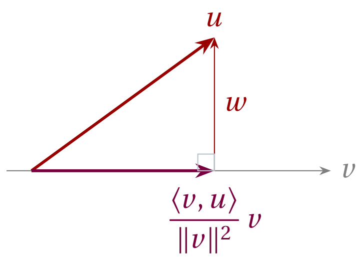

Figure 1: Orthogonal decomposition $u = \frac{\langle v,u\rangle}{\|v\|^2} v + w$ with $w \perp v$. The Pythagorean theorem applied to this right triangle gives Cauchy–Schwarz.

Theorem 7.23 (Cauchy–Schwarz). For all u, v in an inner product space,

$$|\langle u, v\rangle| \leq \|u\|\, \|v\|,$$

with equality if and only if u, v are linearly dependent.

Proof. If $v = 0$ both sides vanish. Otherwise write

$$u = \frac{\langle v, u\rangle}{\|v\|^2}\, v + w, \qquad w = u - \frac{\langle v, u\rangle}{\|v\|^2}\, v.$$

A direct check gives $\langle v, w\rangle = \langle v, u\rangle - \frac{\langle v,u\rangle}{\|v\|^2}\langle v, v\rangle = 0$, so $w \perp v$ (Figure 1). The two pieces are orthogonal, so by the Pythagorean theorem

$$\|u\|^2 = \frac{|\langle v, u\rangle|^2}{\|v\|^2} + \|w\|^2 \geq \frac{|\langle v, u\rangle|^2}{\|v\|^2}.$$

Multiplying by $\|v\|^2$ and taking square roots gives $|\langle v, u\rangle| \leq \|u\|\|v\|$; since $|\langle u, v\rangle| = |\langle v, u\rangle|$, the inequality follows. Equality holds iff $\|w\| = 0$, i.e. $w = 0$, i.e. u is a scalar multiple of v. ∎

Remark 7.24 (The same proof, algebraically). Cauchy–Schwarz is often presented as positivity of a quadratic: for $v \neq 0$ set $\lambda = \langle v, u\rangle / \|v\|^2$ and expand

$$0 \leq \|u - \lambda v\|^2 = \|u\|^2 - \frac{|\langle v, u\rangle|^2}{\|v\|^2},$$

which rearranges to the inequality at once. This is the orthogonal decomposition of Figure 1 seen through algebra rather than geometry; the vector $u - \lambda v$ is exactly the orthogonal part w.

Example 7.25 (Cauchy–Schwarz, checked). For $u = (1, i)$, $v = (2, 1 - i)$ of Example 7.3, $|\langle u, v\rangle| = |1 - i| = \sqrt{2}$, while $\|u\| = \sqrt{1+1} = \sqrt{2}$ and $\|v\| = \sqrt{4+2} = \sqrt{6}$. Indeed $\sqrt{2} \leq \sqrt{2} \cdot \sqrt{6} = 2\sqrt{3}$, with strict inequality because u, v are independent. ◁

Example 7.26 (Orthogonalizing a pair). The decomposition behind the proof gives a recipe: for independent v_1, v_2, the vector $w_2 = v_2 - \frac{\langle v_1, v_2\rangle}{\|v_1\|^2} v_1$ is orthogonal to v_1. For $v_1 = (1, 0)$, $v_2 = (1, 1)$ in $\mathbb{R}^2$ this yields $w_2 = (1, 1) - 1 \cdot (1, 0) = (0, 1)$. Iterating the step over a whole basis is precisely the Gram–Schmidt process of Lecture 8. ◁

Remark 7.27 (Amplitudes are bounded). For unit vectors the Cauchy–Schwarz inequality reads $|\langle \phi \mid \psi\rangle| \leq 1$: no transition amplitude between normalized states exceeds 1 in magnitude, so the Born probability $|\langle \phi \mid \psi\rangle|^2$ is genuinely a probability. The same inequality, applied to operators acting on a state, yields the Heisenberg uncertainty relation (Lecture 9).

Remark 7.28 (Classical inequalities for free). Applied to the function and Frobenius inner products of Examples 7.2 and 7.4, Cauchy–Schwarz immediately yields

$$\left|\int_a^b \overline{f} g\, dt\right|^2 \leq \int_a^b |f|^2\, dt \int_a^b |g|^2\, dt, \qquad |\mathrm{tr}(A^\dagger B)|^2 \leq \mathrm{tr}(A^\dagger A)\ \mathrm{tr}(B^\dagger B).$$

These integral and matrix estimates are awkward to prove head-on but fall out of the one abstract inequality—a payoff of thinking about inner products structurally rather than componentwise.

5 Triangle inequality, parallelogram law, polarization

Theorem 7.29 (Triangle inequality). For all u, v, $\|u + v\| \leq \|u\| + \|v\|$.

Proof. Using $\langle u, v\rangle + \langle v, u\rangle = 2\,\mathrm{Re}\langle u, v\rangle$ and then Cauchy–Schwarz,

$$\|u+v\|^2 = \|u\|^2 + 2\,\mathrm{Re}\langle u, v\rangle + \|v\|^2 \le \|u\|^2 + 2|\langle u, v\rangle| + \|v\|^2 \le \big(\|u\| + \|v\|\big)^2.$$

Taking square roots completes the proof. ■

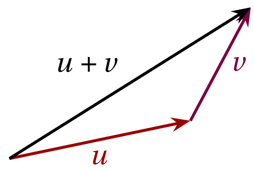

Figure 2: The triangle inequality: the direct path $u+v$ is no longer than the two-leg path of lengths $\|u\|$ and $\|v\|$.

Two further identities express the rigidity of inner-product geometry. Expanding $\|u \pm v\|^2$ and adding or subtracting gives the parallelogram law and the polarization identity.

Proposition 7.30 (Parallelogram law and polarization). For all u, v in a complex inner product space,

$$\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2,$$

and the inner product is recovered from the norm by

$$\langle u, v\rangle = \frac{1}{4} \sum_{\alpha \in \{1,-1,i,-i\}} \alpha\, \|v + \alpha u\|^2.$$

Proof. Expanding $\|u \pm v\|^2 = \langle u \pm v, u \pm v\rangle$ by sesquilinearity gives $\|u \pm v\|^2 = \|u\|^2 + \|v\|^2 \pm 2\Re\langle u, v\rangle$; adding the two sign choices cancels the cross term and gives the parallelogram law. For polarization, the same expansion with $|\alpha| = 1$ gives, for each $\alpha \in \{1, -1, i, -i\}$,

$$\|v + \alpha u\|^2 = \|v\|^2 + \|u\|^2 + \alpha\langle v, u\rangle + \overline{\alpha}\,\langle u, v\rangle.$$

Multiplying by α and summing, the relations $\sum_\alpha \alpha = 0$, $\sum_\alpha \alpha^2 = 0$, and $\sum_\alpha \alpha\overline{\alpha} = 4$ leave $\sum_\alpha \alpha\,\|v + \alpha u\|^2 = 4\langle u, v\rangle$. ■

The parallelogram law characterizes which norms come from an inner product (the p-norms do so only for $p = 2$; the nontrivial converse—that the law forces the polarization formula to be an inner product—is the Jordan–von Neumann theorem, stated without proof), and the polarization identity shows that the inner product carries no information beyond the lengths of vectors. In the real case the polarization identity collapses to $\langle u, v\rangle = \frac{1}{4}\big(\|u+v\|^2 - \|u-v\|^2\big)$, and the angle between vectors is defined by

$$\cos\theta = \frac{\langle u, v\rangle}{\|u\|\,\|v\|},$$

which Cauchy–Schwarz guarantees lies in $[-1, 1]$.

Example 7.31 (An angle and the parallelogram law). In $\mathbb{R}^2$, $u = (1, 0)$ and $v = (1, 1)$ have $\langle u, v\rangle = 1$, $\|u\| = 1$, $\|v\| = \sqrt{2}$, so $\cos\theta = \frac{1}{\sqrt{2}}$ and $\theta = 45°$. The parallelogram law checks: $\|u+v\|^2 + \|u-v\|^2 = \|(2,1)\|^2 + \|(0,-1)\|^2 = 5 + 1 = 6 = 2\|u\|^2 + 2\|v\|^2$. ◁

Example 7.32 (Recovering the product from lengths). For the same real u, v, polarization gives $\langle u, v\rangle = \frac{1}{4}\big(\|u+v\|^2 - \|u-v\|^2\big) = \frac{1}{4}(5-1) = 1$, matching the direct value: the inner product is encoded entirely in the lengths. ◁

Example 7.33 (Distance and the triangle inequality). For $u = (1, i)$ and $v = (2, 1 - i)$ of Example 7.3, the distance is $\|u - v\| = \|(-1,\ 2i - 1)\| = \sqrt{1+5} = \sqrt{6}$, and the triangle inequality $\sqrt{6} \le \|u\| + \|v\| = \sqrt{2} + \sqrt{6}$ holds comfortably. Even in complex space distances obey the familiar geometry. ◁

Example 7.34 (A norm with no inner product). The max norm $\|(x, y)\|_\infty = \max(|x|, |y|)$ on $\mathbb{R}^2$ violates the parallelogram law: for $u = (1, 0)$, $v = (0, 1)$, $\|u + v\|_\infty^2 + \|u - v\|_\infty^2 = 1 + 1 = 2$, but $2\|u\|_\infty^2 + 2\|v\|_\infty^2 = 4$. By Proposition 7.30 no inner product can induce $\|\cdot\|_\infty$: inner-product geometry is genuinely special among norms. ◁

6 States, amplitudes, and what comes next

In quantum mechanics a physical state is a unit vector $|\psi\rangle$ (more precisely a ray, since an overall phase is unobservable). The amplitude $\langle\phi \mid \psi\rangle$ governs interference, and $|\langle\phi \mid \psi\rangle|^2$ is the probability of the outcome $|\phi\rangle$. Orthogonal states ($\langle\phi \mid \psi\rangle = 0$) are perfectly distinguishable: a measurement can never confuse them. The eigenvectors of an observable for distinct eigenvalues will turn out to be orthogonal (Lecture 9), so the possible outcomes of a measurement form an orthogonal family.

Example 7.35 (Interference of qubit states). Let $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. They are orthonormal, since $\langle + \mid - \rangle = \frac{1}{2}(\langle 0 \mid 0\rangle - \langle 1 \mid 1\rangle) = 0$, so a system in $|+\rangle$ is never found in $|-\rangle$. Yet each has overlap $\frac{1}{\sqrt{2}}$ with $|0\rangle$, so a measurement in the $\{|0\rangle, |1\rangle\}$ basis returns either outcome with probability $\frac{1}{2}$. The relative phase distinguishing $|+\rangle$ from $|-\rangle$ is invisible to that measurement but decisive in others—the algebra of interference. ◁

Remark 7.36 (Cauchy–Schwarz and uncertainty). Cauchy–Schwarz is the engine of the Heisenberg uncertainty relation. For observables A, B and a state $|\psi\rangle$, set $\Delta A = A - \langle A\rangle I$ and $\Delta B = B - \langle B\rangle I$. Cauchy–Schwarz applied to the vectors $\Delta A|\psi\rangle$ and $\Delta B|\psi\rangle$ gives

$$\langle(\Delta A)^2\rangle\,\langle(\Delta B)^2\rangle \ge |\langle\psi|\,\Delta A\,\Delta B\,|\psi\rangle|^2 \ge \left(\tfrac{1}{2}|\langle[A, B]\rangle|\right)^2.$$

With $[\hat{x}, \hat{p}] = i\hbar I$ this becomes $\Delta x\,\Delta p \ge \hbar/2$: a bound on amplitudes turns into a bound on simultaneous knowledge. The full argument waits for adjoints in Lecture 9.

Remark 7.37 (Toward Hilbert space). A complex inner product space that is also *complete*—every Cauchy sequence of vectors converges to a vector in the space—is a *Hilbert space*. Every finite-dimensional inner product space is automatically complete, so the distinction is invisible here; the infinite-dimensional state spaces of quantum mechanics, such as L^2, require completeness as a genuine extra axiom, taken up in Lecture 13.

Example 7.38 (Closest point on a line). Among all scalar multiples of a unit vector e, the one closest to a given u is $\langle e, u\rangle\, e$: the error $u - \langle e, u\rangle\, e$ is orthogonal to e, so by Pythagoras $\|u - ce\|^2 = \|u - \langle e, u\rangle\, e\|^2 + |c - \langle e, u\rangle|^2$ is smallest at $c = \langle e, u\rangle$. This orthogonal projection is the principle behind least-squares fitting and the orthonormal expansions of Lecture 8. ◁

That raises the question driving the next lecture: can we always choose a basis of mutually orthogonal unit vectors—an *orthonormal basis*—and expand any state in it? The answer is yes, by the Gram–Schmidt procedure, and in such a basis the inner product becomes the familiar sum $\sum_k \overline{u_k} v_k$ and operators acquire especially clean matrices.

Summary of main concepts

An inner product is sesquilinear (conjugate-linear in the first slot, linear in the second, in the physics convention), conjugate-symmetric, and positive definite; in Dirac notation

$\langle \phi \mid \psi \rangle$ is linear in the ket and conjugate-linear in the bra. It induces a norm $\|v\| = \sqrt{\langle v, v \rangle}$, orthogonality $\langle u, v \rangle = 0$, and the Pythagorean theorem. The Cauchy–Schwarz inequality $|\langle u, v \rangle| \leq \|u\|\|v\|$ follows from orthogonal decomposition and bounds every amplitude; from it come the triangle inequality and, with the parallelogram law and polarization identity, the full geometry of the space. In physics, states are unit vectors, $|\langle \phi \mid \psi \rangle|^2$ is a probability, and orthogonality means distinguishability—the setting in which the spectral theorem will live.

- **Inner product (physics convention)**—sesquilinear, conjugate-symmetric, positive-definite; $\langle \phi \mid \psi \rangle$ is linear in the ket.
- **Norm and orthogonality**—$\|v\| = \sqrt{\langle v, v \rangle}$; $\langle u, v \rangle = 0$; the Pythagorean theorem.
- **Cauchy–Schwarz**—$|\langle u, v \rangle| \leq \|u\| \, \|v\|$; bounds every amplitude.
- **Triangle inequality, parallelogram law, polarization**—the metric geometry of the space; recover $\langle \cdot, \cdot \rangle$ from $\|\cdot\|$.
- **Physics picture**—states are unit vectors, $|\langle \phi \mid \psi \rangle|^2$ a probability, orthogonality means distinguishability.

Exercises for this lecture are in the companion sheet `exercises-7.pdf`.